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(Printed Pages 4)
Roll No. _____

24/1162

B.C.A. (Fifth Semester)

Examination, 2024

First Paper

(Introduction to DBMS)

Time : Two Hours]

[Maximum Marks : 75

Note : Attempt **all** sections as directed.

Section-A

(Very Short Answer Type Questions)

Note: Attempt all the 05 (five) questions.

Each question carries 02 (two) marks
and answer of each question should not
exceed 50 words. $5 \times 2 = 10$

1. (a) What is the role of integrity constraints?
(b) Explain the concept recovery.
(c) What do you mean by normalize the data?

P.T.O.

24/1162

- (d) Under what conditions must a foreign key not be well?
- (e) Define user defined functions in SQL.

Section-B

(Short Answer Type Questions)

Note : Attempt any **05 (five)** questions out of total 08 (Eight) questions. Each question carries 05 (five) marks and answer of each question should not exceed 100 words. $5 \times 5 = 25$

2. (a) Explain the terms: primary key and candidate key.
(b) What is the difference between specialization and generalisation.
(c) What is log files? How it is maintained?
(d) What is functional dependency? Explain the role of functional dependency in the process of normalization.

24/1162

- (e) Discuss various locking techniques used in the concurrently control.
- (f) Explain "ACID" properties of transaction in detail.
- (g) How does a check point mechanism help in database recovery?
- (h) Explain the responsibilities of a database Administrator.

Section-C

(Long Answer Type Questions)

Note : Attempt any **two (02)** out of total 04 (four) questions. Each question carries 20 (twenty) marks and answer of each question should not exceed 400 words.

2×20=40

3. (a) What do you understand by Relational model of database system? Elaborate the major characteristics of relational database management system.

3

P.T.O.

24/1162

- (b) What do you understand by the term "Normalization" in DBMS? Write statement for third normal form (3NF) and discuss the insert, delete and update anomalies associated with 3NF, with the help of suitable example.
- (c) What is database modeling? How is E-R model different from other data models? What are the main advantages of the E-R model?
- (d) Explain ANSI-SPARC 3 Level architecture of DBMS and type of data independence between different levels give the suitable diagram in support of your explanation.

4